

AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

- **Problem Statement ID** – SIH26090
- **Problem Statement Title-** AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans
- **Organisation:** Ministry of Social Justice and Empowerment (MoSJE)
- **Theme** - Miscellaneous
- **PS Category-** Software
- **Team ID-** 126401u
- **Team Name** – Team Zero Click



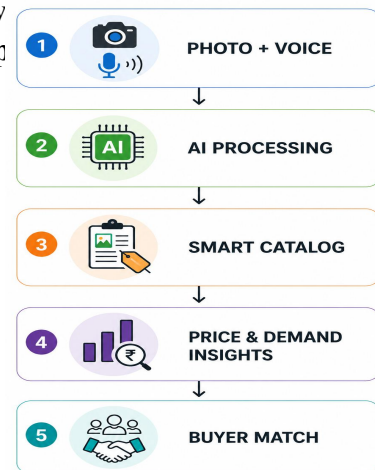
IDEA TITLE

IDEA/SOLUTION:

CraftLink AI — a “virtual business manager” for artisans who have never sold online. One photo + one spoken sentence becomes a market-ready listing.

- **Photo-based product capture**
Background removal, lighting fix, e-commerce framing.
- **Voice-first input**
Local-language speech-to-text — no typing, no English.
- **Automatic catalog generation**
SEO title, description, category and keywords.
- **Evidence-based pricing** Materials + labour + market bands + signals read from the photo and description — every factor shown.
- **Buyer & market matching** ONDC/GeM-ready listing

fields; bulk enquiries from B2B and gov app



Target Users: Rural & marginalized artisans

PROBLEM RESOLUTION:

- **No professional photos** — AI removes cluttered backgrounds, corrects lighting, formats to e-commerce standards.
- **No English, low digital literacy** — local-language voice notes become SEO-friendly descriptions; zero typing.
- **No pricing knowledge** — ML reads the photo and the description alongside materials, labour and market trends, and shows every reason.
- **Only seasonal melas** — Dilli Haat, Surajkund and Shilp Samagam give weeks a year; this gives 365 days.
- **Middlemen take the margin** — buyers raise bulk enquiries and the artisan quotes back directly, in their own language.

UNIQUE VALUE PROPOSITION (UVP):

- **Voice-first, local-language cataloging** — photo + spoken sentence → publishable listing in minutes.
- **Evidence-based pricing, not a black box** — the artisan sees each factor, including what the photo added, and can override it.
- **Runs without the internet or an API key** — Whisper and NLLB-200 do the language work on-device/on-server.
- **Human-in-the-loop by design** — nothing publishes until the artisan approves the AI's words and price.

KEY DIFFERENTIATOR

Not another marketplace — an AI co-seller that does the photography, copywriting and pricing an artisan cannot, runs offline in nine languages, and hands the finished listing to the marketplaces that already exist.

TECHNOLOGIES USED:



AI & ML:

- **Computer Vision** — rembg, OpenCV
- **Speech-to-Text** — faster-whisper (offline)
- **NLLB-200** — 9-language listings
- **Scikit-learn** — price band model



FRONTEND:

- React + Vite + Tailwind
- Installable PWA + Android APK (Capacitor)



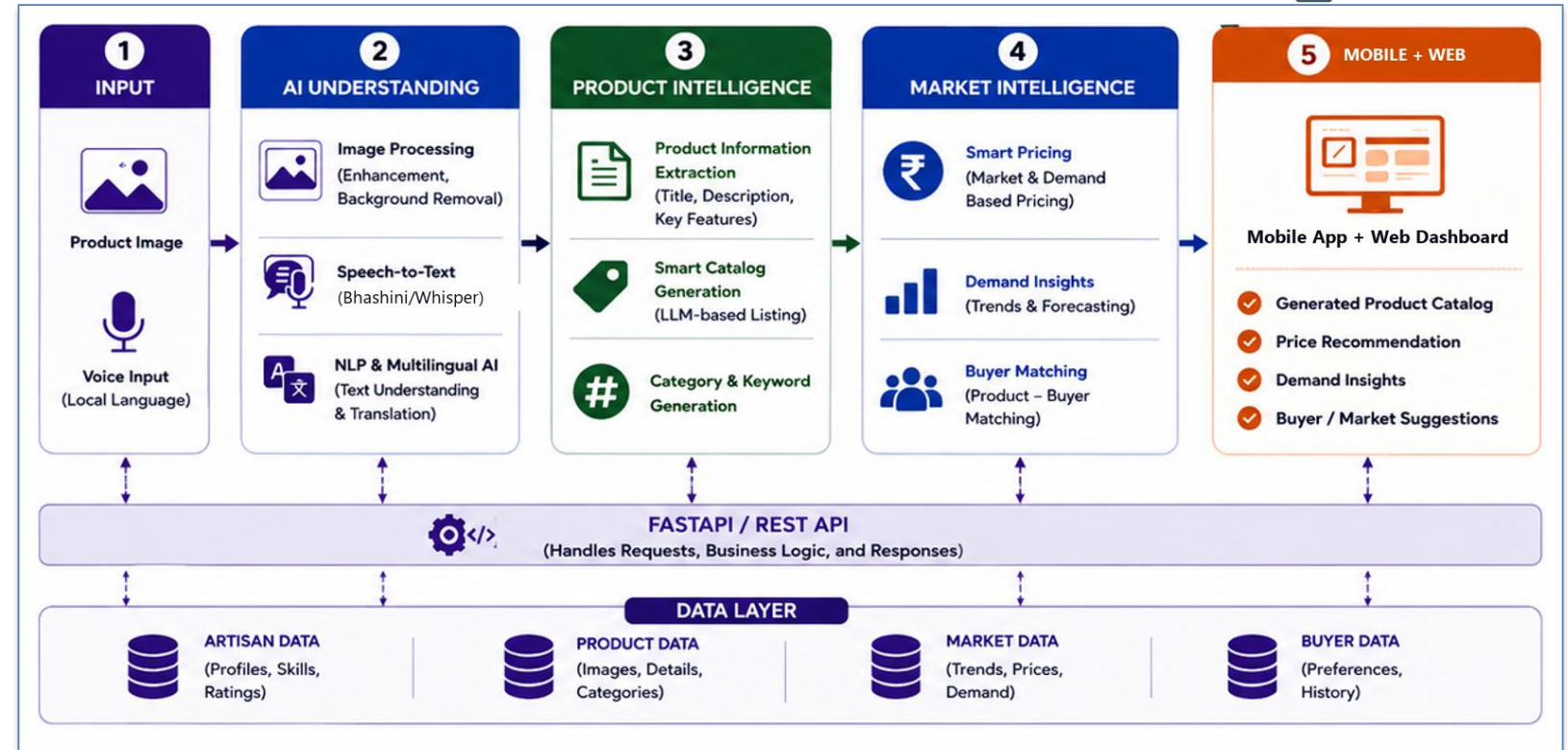
BACKEND & DATABASE:

- Python + FastAPI
- SQLite / PostgreSQL + SQLAlchemy



INTEGRATIONS:

- ONDC/GeM-ready catalogue fields
- REST APIs + JSON
- .env — secure configuration



HOW IT DELIVERS VALUE:



CORE INNOVATION:

- From craft to commerce — one offline AI pipeline turns a photo + a spoken sentence into a market-ready listing in 9 languages, prices it from the photo and the description, and opens it to bulk buyers.

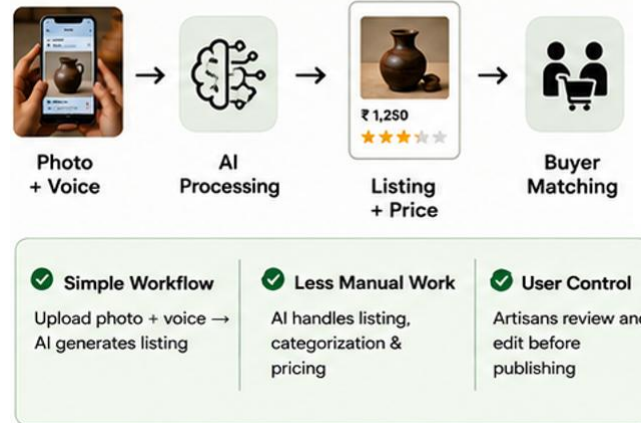


TECHNICAL FEASIBILITY:

- ✓ **Integration:** Computer Vision, Whisper ASR, NLLB-200 translation and ML automate photo clean-up, listing and pricing.
- ✓ **Web + Mobile Architecture:** React + FastAPI, shipped as a signed Android APK and an offline-capable PWA — no specialised hardware.
- ✓ **Multilingual Interaction:** Voice input and full UI — including fonts — in 9 Indian languages; no typing, no English.
- ✓ **Proven Stack:** Every component is open-source and already running in our working prototype.



OPERATIONAL FEASIBILITY



ECONOMIC VIABILITY:

- ✓ **Zero cost to the artisan** → no listing fee and no commission; the artisan keeps the margin.
- ✓ **Low-cost deployment** → every model runs offline on our own server; no per-call API bill.
- ✓ **Publicly fundable** → fits MoSJE corporation (NSFDC / NBCFDC / NSKFDC) and CSR funding routes.
- ✓ **Sustainable expansion** → new languages, crafts and markets added without infrastructure change.



CHALLENGES:

- ✓ **Image Quality** → Poor product images may reduce AI accuracy.
- ✓ **Voice Accuracy** → Accents and local-language variations may affect transcription.
- ✓ **AI Reliability** → Generated product information may require human correction.
- ✓ **Market Data** → Limited or changing market data can affect price and demand insights.
- ✓ **Digital Adoption** → Some artisans may have limited experience with online platforms.



MITIGATION STRATEGIES:

- ✓ **Image Preprocessing** → Improve image quality before AI analysis.
- ✓ **Voice Processing** → a fast Whisper pass with a large-v3 fallback when confidence drops.
- ✓ **Human Verification** → Allow artisans to review and edit AI-generated details.
- ✓ **Data Validation** → Cross-check market trends before generating pricing recommendations.
- ✓ **Simple User Interface** → Keep the workflow easy for users with limited digital experience.



SOLUTION POTENTIAL:

- **Working build today** — photo → voice → 9-language listing → price → order → bulk quote, end to end, offline.
- **Phase 1 (built):** Android APK + installable PWA, 9 languages, offline-capable.
- **Phase 2 (3–9 months):** Bhashini APIs for the remaining scheduled languages; certified ONDC seller onboarding.
- **Phase 3:** Scale through NSFDC, NBCFDC and NSKFDC artisan networks.



IMPACT ON STAKEHOLDERS:

- 1. MoSJE BENEFICIARY ARTISANS** (SC, OBC, DNT, Safai Karamcharis, PwD)
→ **Year-round digital market access** without English, a camera or digital skills.
- 2. BUYERS & CUSTOMERS**
→ **Authentic, discoverable crafts** with GI-aware categories and honest, explained pricing.
- 3. ARTISAN COMMUNITIES & CLUSTERS**
→ **Craft and livelihood preservation** — traditional skills gain visibility and commercial value locally.
- 4. MoSJE & ITS CORPORATIONS** (NSFDC / NBCFDC / NSKFDC)
→ **A measurable digital channel** extending Bharat-TULIP beyond seasonal melas.



OUR MISSION:

“To empower India’s artisans with AI-driven, language-inclusive tools that turn traditional craftsmanship into market-ready products, enabling wider market access, fairer opportunities, and sustainable livelihoods.”



BENEFITS OF OUR SOLUTION:

- **Higher Market Reach**
A 365-day online storefront instead of a few exhibition weeks a year.
- **Better Earning Potential**
Direct buyer linkage removes the middleman cut — today up to 60% of product value.
- **Language-Inclusive Commerce** Voice, screen and listings in 9 Indian languages today — Hindi, Bengali, Marathi, Telugu, Tamil, Gujarati, Kannada, Malayalam, English.
- **Faster Product Onboarding**
Photo + voice → publishable listing in minutes, not a day of form-filling.
- **Smarter Buyer–Artisan Matching**
Categories, GI tags and demand trends drive discovery.

From making products to reaching markets — one platform removes the digital barriers between India’s ~7 million artisans and their buyers.

REAL-WORLD OUTCOMES:



1. Ministry of Social Justice and Empowerment, Government of India.

Bharat-TULIP (Traditional Artisans' Upliftment Livelihood Programme) — e-marketplace for marginalized artisans; launched 5 Nov 2024, 1,500+ artisans onboarded.

<https://nbcfdc.gov.in/nbcfdc/web/tulip>

2. Ministry of Social Justice and Empowerment, Government of India.

PM-DAKSH — skilling and livelihood support for SC, OBC, EWS, DNT and Safai Karamchari artisans. <https://socialjustice.gov.in/>

3. Development Commissioner (Handicrafts), Ministry of Textiles, Government of India.

Crafts Clusters of India — artisan development, marketing and cluster support; India Handmade marketplace. <https://handicrafts.nic.in/>

4. Bhashini — National Language Translation Mission, MeitY, Government of India.

ASR, machine translation and TTS across the 22 scheduled Indian languages (300+ models). <https://bhashini.gov.in/>

5. Open Network for Digital Commerce (ONDC), DPIIT-backed initiative.

Seller Network Participant onboarding, digital cataloguing and marketplace discovery. <https://www.ondc.org/>

6. Intellectual Property India, Government of India.

Geographical Indications (GI) — protection of handicrafts and traditional products (318 GI-tagged crafts). <https://ipindia.gov.in/>

7. India Brand Equity Foundation (IBEF), Ministry of Commerce & Industry.

Indian Handicrafts Industry — ~7 million artisans; handicraft exports ₹33,122 crore in FY 2024-25. <https://www.ibef.org/exports/handicrafts-industry-india>

DESIGNED TO PLUG INTO INDIA'S DIGITAL PUBLIC INFRASTRUCTURE

Built to plug into ONDC — market reach · Bharat-TULIP (MoSJE) — artisan marketplace · GI Registry — craft authenticity · PM-DAKSH — artisan skilling · Bhashini — roadmap for the remaining scheduled languages